

Calculus II

Definite integrals of the form $\int_p^q \sqrt{ax^2 + c} dx,$
 $a, c > 0$

Todor Milev

2019

Example

Find the area locked b-n the hyperbolas $y = \pm\sqrt{x^2 + 1}$ and $x = \pm 2\sqrt{2}$.

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$$\text{where } \begin{cases} u = \frac{\sqrt{2}}{2}(y - x) \\ v = \frac{\sqrt{2}}{2}(y + x) \end{cases}.$$

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Signed distance b-n (x, y) and line $u = 0$ equals u . Similarly compute that signed distance b-n (x, y) and the line $v = 0$ equals v .
 $\Rightarrow y^2 - x^2 = 1$ is the **hyperbola**
 $v = \frac{1/2}{u}$ in the (u, v) -plane.

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The area in question is:

$$\int_{\text{?}}^{\text{?}} 2\sqrt{x^2 + 1} dx$$

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$$\begin{aligned} & \int_{-2\sqrt{2}}^{2\sqrt{2}} 2\sqrt{x^2 + 1} dx \\ &= \left[x\sqrt{x^2 + 1} \right. \\ & \quad \left. + \ln \left(\sqrt{x^2 + 1} + x \right) \right]_{-2\sqrt{2}}^{2\sqrt{2}} \end{aligned}$$

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The area in question is:

$$\begin{aligned}
 & \int_{-2\sqrt{2}}^{2\sqrt{2}} 2\sqrt{x^2 + 1} dx \\
 &= 2 \left[x\sqrt{x^2 + 1} \right. \\
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 & \quad \left. + \ln \left(\sqrt{x^2 + 1} + x \right) \right]_0^{2\sqrt{2}} \\
 &= 2 \left(2\sqrt{2}\sqrt{(2\sqrt{2})^2 + 1} \right.
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 &= 2 \left(2\sqrt{2}\sqrt{(2\sqrt{2})^2 + 1} \right. \\
 & \quad \left. + \ln \left(\sqrt{(2\sqrt{2})^2 + 1} + 2\sqrt{2} \right) \right)
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 &= 2 \left[x\sqrt{x^2 + 1} \right. \\
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 &= 2 \left(2\sqrt{2}\sqrt{(2\sqrt{2})^2 + 1} \right. \\
 & \quad \left. + \ln \left(\sqrt{(2\sqrt{2})^2 + 1} + 2\sqrt{2} \right) \right) \\
 &= 12\sqrt{2} + 2\ln \left(3 + 2\sqrt{2} \right)
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 & \quad \left. + \ln \left(\sqrt{x^2 + 1} + x \right) \right]_0^{2\sqrt{2}} \\
 &= 2 \left(2\sqrt{2}\sqrt{(2\sqrt{2})^2 + 1} \right. \\
 & \quad \left. + \ln \left(\sqrt{(2\sqrt{2})^2 + 1} + 2\sqrt{2} \right) \right) \\
 &= 12\sqrt{2} + 2\ln(3 + 2\sqrt{2}) \\
 &\approx 20.496
 \end{aligned}$$

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The area in question is:

- Recall: integral can be solved via $x = \tan \theta$.

$$\begin{aligned}
 & \int_{-2\sqrt{2}}^{2\sqrt{2}} 2\sqrt{x^2 + 1} dx \\
 &= 2 \left[x\sqrt{x^2 + 1} + \ln \left(\sqrt{x^2 + 1} + x \right) \right]_{-2\sqrt{2}}^{2\sqrt{2}} \\
 &= 2 \left(2\sqrt{2}\sqrt{(2\sqrt{2})^2 + 1} + \ln \left(\sqrt{(2\sqrt{2})^2 + 1} + 2\sqrt{2} \right) \right) \\
 &= 12\sqrt{2} + 2 \ln \left(3 + 2\sqrt{2} \right) \\
 &\approx 20.496
 \end{aligned}$$

Example

Find the area locked b-n the hyperbolas $y = \pm\sqrt{x^2 + 1}$ and $x = \pm 2\sqrt{2}$.
The area in question is:

- Recall: integral can be solved via $x = \tan \theta$.
- Geometric interpretation of θ ?

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